

第 3 周实验报告 (Packaging and Shipping Code / 智能体编程 / 不止于代码 / Python 与 PyTorch)

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1 第 9 题从源码构建并在干净环境安装 Wheel

主题: Packaging and Shipping Code

按 src layout 建最小命令行包: `__init__.py` 标记包, `cli.py` 用 `argparse` 解析 `--name`, `pyproject.toml` 声明构建后端与入口点 `sdt-greet`。

```
# src/greetlab/cli.py
import argparse

def main():
    p = argparse.ArgumentParser()      # 解析器
    p.add_argument("--name", required=True) # 必填 --name
    a = p.parse_args()                 # 解析 argv
    print(f"Hello, {a.name}!")         # 插值输出
```

```
# pyproject.toml
[build-system]
requires = ["setuptools>=68"]
build-backend = "setuptools.build_meta"

[project]
name = "greetlab-25020007111"
version = "0.1.0"

[project.scripts]
sdt-greet = "greetlab.cli:main"
```

```
mkdir q09 && cd q09
mkdir -p src/greetlab
pip install build
python -m build          # 产出 dist/greetlab_25020007111-0.1.0-py3-none-any.whl

python -m venv .venv     # 干净虚拟环境
source .venv/bin/activate
```

```

pip install dist/greetlab_25020007111-0.1.0-py3-none-any.whl

cd ..                                # 切到 q09 之外, 证明入口点已全局安装
sdt-greet --name 25020007111 # Hello, 25020007111!
which sdt-greet                  # 指向 .venv/bin/sdt-greet

```

```

1.0-py3-none-any.whl' and adding 'build/bdist.linux-x86_64/wheel' to it
adding 'greetlab/__init__.py'
adding 'greetlab/cli.py'
adding 'greetlab_25020007111-0.1.0.dist-info/METADATA'
adding 'greetlab_25020007111-0.1.0.dist-info/WHEEL'
adding 'greetlab_25020007111-0.1.0.dist-info/entry_points.txt'
adding 'greetlab_25020007111-0.1.0.dist-info/top_level.txt'
adding 'greetlab_25020007111-0.1.0.dist-info/RECORD'
removing build/bdist.linux-x86_64/wheel
Successfully built greetlab_25020007111-0.1.0.tar.gz and greetlab_25020007111-0.1.0-py3-none-any.whl
tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ ls dist
greetlab_25020007111-0.1.0-py3-none-any.whl  greetlab_25020007111-0.1.0.tar.gz
tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ ls -R
.:
dist  pyproject.toml  src

./dist:
greetlab_25020007111-0.1.0-py3-none-any.whl  greetlab_25020007111-0.1.0.tar.gz

./src:
greetlab  greetlab_25020007111.egg-info

./src/greetlab:
cli.py  __init__.py  __pycache__

./src/greetlab/__pycache__:
cli.cpython-314.pyc  __init__.cpython-314.pyc

./src/greetlab_25020007111.egg-info:
dependency_links.txt  entry_points.txt  PKG-INFO  SOURCES.txt  top_level.txt
tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ python -m venv .venv

```

图 1: 第 9 题: python -m build 构建成功, dist/ 下生成 .whl 与 .tar.gz, ls -R 可见 src 布局

```

tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ python -m venv .venv
tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ source .venv/bin/activate
(.venv) tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ which python
~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/q09/.venv/bin/python
(.venv) tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ pip install dist/greetlab_25020007111-0.1.0-py3-none-any.whl
Processing ./dist/greetlab_25020007111-0.1.0-py3-none-any.whl
greetlab-25020007111 is already installed with the same version as the provided wheel. Use --force-reinstall to force an installation
of the wheel.

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
(.venv) tancg@fedora:~/../sys-dev-tools-course-2026/week3/q09$ cd ..
(.venv) tancg@fedora:~/../git_repo/sys-dev-tools-course-2026/week3$ sdt-greet --name 25020007111
Hello, 25020007111!
(.venv) tancg@fedora:~/../git_repo/sys-dev-tools-course-2026/week3$ which sdt-greet
~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/q09/.venv/bin/sdt-greet
(.venv) tancg@fedora:~/../git_repo/sys-dev-tools-course-2026/week3$
~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3

```

图 2: 第 9 题: 新建 venv、which python 指向 venv, 从 wheel 安装后切到 q09 之外运行 sdt-greet --name 25020007111 输出 Hello

解题感悟

1. src layout 把源码放进 src/, 配合 __init__.py 标记包, 避免误 import 未构建的半成品。
2. pyproject.toml 是现代打包的单一事实来源: [build-system] 指定 setuptools 后端, [project.scripts] 生成命令行入口点。

3. `python -m build` 是 PEP 517 构建前端，产出纯 Python wheel (py3-none-any) 与 sdist。
4. 关键验证：在 q09 之外运行 `sdt-greet`，才真正证明 wheel 装好、入口点生效，而非凑巧读到本地源码。

2 第 10 题让编程智能体进入可验证的修复循环

主题：智能体编程

复制 q09 为 q10，先写一个会失败的测试 (TDD): `--name` 只含空白时应以退出码 2 结束。再让智能体修改实现并运行测试，最后人工 `git diff` 审改动、复跑测试。

```
# test_cli.py
import sys
import pytest
from greetlab.cli import main

def test_blank_name_exits_2():
    with pytest.raises(SystemExit) as excinfo:
        sys.argv = ["sdt-greet", "--name", " "] # 空白 name
        main()
    assert excinfo.value.code == 2
```

```
cp -r q09 q10 && cd q10
PYTHONPATH=src pytest # 先确认失败: DID NOT RAISE SystemExit
```

给智能体的提示 (目标 + 约束 + 验证命令):

目标：修改 `q10/src/greetlab/cli.py` 的 `main`，使 `--name` 只含空白时以 `SystemExit(2)` 结束。
约束：不要改动 `test_cli.py` 和 `pyproject.toml`；不要删除 `--name` 参数。
验证命令：`cd q10 && PYTHONPATH=src pytest`
请修改实现并运行测试，直到通过。

```
# src/greetlab/cli.py (修复后)
def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    if not a.name.strip(): # 只含空白判为非法
        raise SystemExit(2)
    print(f"Hello, {a.name}!")
```

```
git diff # 人工检查: 确认只改了 cli.py
git checkout -- <无关文件> # 撤销智能体的无关修改
PYTHONPATH=src pytest # 1 passed
```

智能体会话记录 (share): <https://opncl.ai/share/45fy5UJ0>

```
week3 > q10 >  test_cli.py >  test_blank_name_exits_2
```

```
1  # test_cli.py
2  import sys
3  import pytest
4  from greetlab.cli import main
5
6  def test_blank_name_exits_2():
7      # 传入只含空白字符的 name, main 应以退出码 2 结束
8      with pytest.raises(SystemExit) as excinfo:
9          sys.argv = ["sdt-greet", "--name", "   "]
10         main()
11     assert excinfo.value.code == 2
```

图 3: 第 10 题: VS Code 中新建 `test_cli.py`, `sys.argv = ["sdt-greet", "--name", " "]`, 断言退出码为 2

```
tangc@fedora:~/sys-dev-tools-course-2026/week3/q10$ pytest
===== test session starts =====
platform linux -- Python 3.14.7, pytest-8.4.2, pluggy-1.6.0
rootdir: /home/tangc/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/q10
configfile: pyproject.toml
plugins: anyio-4.13.0
collected 1 item

test_cli.py F [100%]

===== FAILURES =====
----- test_blank_name_exits_2 -----
def test_blank_name_exits_2():
    # 传入只含空白字符的 name, main 应以退出码 2 结束
    > with pytest.raises(SystemExit) as excinfo:
        ~~~~~
E       Failed: DID NOT RAISE <class 'SystemExit'>

test_cli.py:8: Failed
----- Captured stdout call -----
Hello, !
===== short test summary info =====
FAILED test_cli.py::test_blank_name_exits_2 - Failed: DID NOT RAISE <class 'SystemExit'>
===== 1 failed in 0.06s =====
tangc@fedora:~/sys-dev-tools-course-2026/week3/q10$
```

图 4: 第 10 题: 先跑测试确认失败——DID NOT RAISE SystemExit, 捕获输出为 Hello, !



图 5: 第 10 题: 智能体按目标修改 cli.py, 新增 if not a.name.strip(): raise SystemExit(2), 并运行测试通过

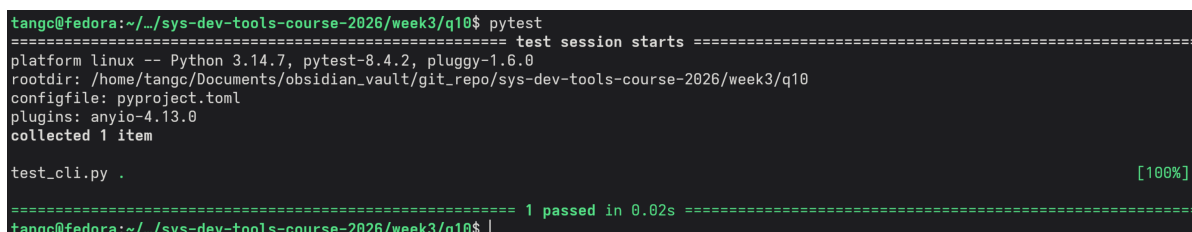


图 6: 第 10 题: 修复后 pytest 显示 1 passed

解题感悟

1. TDD: 先写失败的测试描述正确行为, 再改实现让它通过, 目标明确、可自动验证。
2. 给智能体三点: 目标 (改什么行为)、约束 (不许动什么)、验证命令 (怎么算对) —— 形成可验证修复循环。
3. argparse 对参数错误约定退出码 2; p.error(...) 会打印 usage 再退出 2, 比裸 sys.exit(2) 更友好。
4. 人必须把关: git diff 审改动、撤销无关修改、复跑测试; 智能体可能“过拟合”测试。

3 第 11 题把“无法处理”的协作材料改成可执行信息

主题：不止于代码

围绕“空姓名仍输出问候语”缺陷，把三段质量差的协作材料改写为可执行信息：Issue 补环境/复现/期望/实际，提交信息用祈使语气标题加“为什么 + 怎么做”，评审意见给具体行为 + 风险 + 建议并标注 Blocking/Suggestion/Nit。

```
# communication.md
```

```
## Issue (重写)
```

标题：sdt-greet 对空白 --name 仍输出问候并以 0 退出

环境：Linux / Python 3.13 (Windows 是否可复现待确认)

复现：sdt-greet --name " "

期望结果：以非零退出码结束，不输出问候语。

实际结果：输出 "Hello, !" 并以退出码 0 结束。

```
## 提交信息 (重写)
```

标题：Validate --name to reject blank names

正文：--name 只含空白时仍输出问候并以 0 退出，调用方无法判错。

在 parse_args 后校验 a.name.strip()，为空则 raise SystemExit(2)。

```
## 评审意见 (重写)
```

[Blocking] 新增校验缺少对应测试，需补空白与空串两个用例。

[Nit] 错误消息建议统一格式，便于断言。

[Suggestion] 可复用 argparse type 校验，逻辑更内聚。



图 7: 第 11 题: VS Code 编辑 communication.md (左) 与预览 (右), Issue/提交信息/评审意见三段均按可执行标准重写

解题感悟

1. 好的 Issue: 标题一句话说清问题, 正文含环境、复现命令、期望 vs 实际, 未知写“待确认”。

2. 好的提交信息：祈使语气标题即摘要，正文分“问题 + 解决方案”。
3. 好的评审意见：指出具体行为 + 风险 + 建议动作，并区分 Blocking / Suggestion / Nit。
4. 共同原则：可执行、专业、简洁、不含情绪，全文控制在 400 字以内。

4 第 12 题修复一个可复现的线性回归训练循环

主题：Python 与 PyTorch

补全训练循环：`zero_grad` → `backward` → `step`；训练后 `model.eval()` 并在 `torch.no_grad()` 中计算最终损失，打印 `loss/weight/bias`。不得直接赋值 `weight/bias`。

```
# q12/train.py
import torch
from torch import nn

torch.manual_seed(20260907)
x = torch.linspace(-1, 1, 101).reshape(-1, 1)
y = 3 * x - 1

model = nn.Linear(1, 1)
loss_fn = nn.MSELoss()
opt = torch.optim.SGD(model.parameters(), lr=0.1)

for _ in range(200):
    pred = model(x)
    loss = loss_fn(pred, y)
    opt.zero_grad()      # TODO01: 清空梯度（不清会累加）
    loss.backward()      # TODO02: 反向传播求梯度
    opt.step()           # TODO03: 按梯度更新参数

model.eval()            # 评估模式
with torch.no_grad():   # 不计算梯度
    pred = model(x)
    final_loss = loss_fn(pred, y)
print(f"final loss = {final_loss.item():.6f}")
print(f"weight = {model.weight.item():.6f}")
print(f"bias = {model.bias.item():.6f}")
```

```
python q12/train.py
# final loss = 0.000000 (< 0.001, 收敛)
# weight = 2.999997 (~= 3)
# bias = -1.000000 (~= -1)
```

```
train.py U X
week3 > q12 > train.py > ...
1  # q12/train.py
2  import torch
3  from torch import nn
4
5  torch.manual_seed(20260907)
6  x = torch.linspace(-1, 1, 101).reshape(-1, 1)
7  y = 3 * x - 1
8
9  model = nn.Linear(1, 1)
10 loss_fn = nn.MSELoss()
11 opt = torch.optim.SGD(model.parameters(), lr=0.1)
12
13 for _ in range(200):
14     pred = model(x)                # 前向：算出预测
15     loss = loss_fn(pred, y)         # 算损失
16     opt.zero_grad()                # 清空梯度（不清会累加）
17     loss.backward()                 # 反向传播求梯度
18     opt.step()                     # 按梯度更新参数
19
20 model.eval()                       # 切到评估模式
21 with torch.no_grad():              # 不计算梯度
22     pred = model(x)
23     final_loss = loss_fn(pred, y)   # 最终损失
24 print(f"final loss = {final_loss.item():.6f}")
25 print(f"weight = {model.weight.item():.6f}")
26 print(f"bias = {model.bias.item():.6f}")
```

图 8: 第 12 题: 补全 train.py, 训练三步与评估阶段注释齐全

```
tangc@fedora:~/sys-dev-tools-course-2026/week3/q11$ cd ..
tangc@fedora:~/git_repo/sys-dev-tools-course-2026/week3$ python q12/train.py
final loss = 0.000000
weight = 2.999997
bias = -1.000000
tangc@fedora:~/git_repo/sys-dev-tools-course-2026/week3$
```

图 9: 第 12 题: 运行输出 final loss = 0.000000、weight = 2.999997、bias = -1.000000, 训练收敛

解题感悟

1. PyTorch 训练三步: `opt.zero_grad()` 清梯度 (梯度默认累加)、`loss.backward()` 反向传播、`opt.step()` 更新参数, 顺序不能乱。
2. 评估用 `model.eval()` + `torch.no_grad()`, 关闭训练专用行为并省内存。
3. 数据由 $y = 3x - 1$ 生成, 故收敛后 $\text{weight} \approx 3$ 、 $\text{bias} \approx -1$; 必须靠训练学出来, 不能直接赋值。
4. `torch.manual_seed(20260907)` 固定随机种子, 保证结果可复现。

5 课后练习 (missing-semester: shipping-code / agentic-coding / beyond-code)

5.1 A1. printenv 保存环境 → 建 venv → diff 前后差异

```

printenv > before.txt      # 环境快照
python -m venv .venv
source .venv/bin/activate
printenv > after.txt
diff before.txt after.txt  # 新增 VIRTUAL_ENV、PS1, PATH 开头插入 .venv/bin

which deactivate           # 无输出 (不是可执行文件)
type deactivate            # deactivate 是一个 bash 函数

```

激活后 PATH 开头被插入 `.venv/bin`，命令查找按 PATH 从前往后，故 `python/pip` 优先命中 `venv`；`deactivate` 是 `activate` 脚本定义的函数，负责还原 PATH、VIRTUAL_ENV。

```
tangc@fedora:~/../git_repo/sys-dev-tools-course-2026/week3$ mkdir ./ex && cd ./ex
tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ printenv > before.txt
tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ echo "$PATH"
/home/tangc/.local/bin:/home/tangc/bin:/usr/local/bin:/usr/bin
tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ python -m venv .venv
tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ source .venv/bin/activate
(.venv) tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ echo "$PATH"
/home/tangc/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/.venv/bin:/home/tangc/.local/bin:/home/tangc/bin:/usr/local/bin:/usr/bin
(.venv) tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ printenv > after.txt
(.venv) tangc@fedora:~/../sys-dev-tools-course-2026/week3$ ex$ diff before.txt after.txt
30a31
> VIRTUAL_ENV=/home/tangc/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/.venv
47a49
> VIRTUAL_ENV_PROMPT=.venv
49a52
> PS1=\[\e\]133;k;start_kitty\a\]\[\e\]133;D;$?\a\e\]133;A;a\]\[\e\]133;k;end_kitty\a\](.venv) ${PROMPT_START@P}\[\e\0m\]${PROMPT_HIGHLIGHT:~}\[\e\]${PROMPT_HIGHLIGHT}m\]\[${PROMPT_COLOR:+\[\e\]${PROMPT_COLOR}m\}\]${PROMPT_CONTAINER}${PROMPT_USERHOST:+${PROMPT_USERHOST@P}${PROMPT_SEPARATOR:+\[\e\0m\]${PROMPT_SEPARATOR_COLOR:+\[\e\]${PROMPT_SEPARATOR_COLOR}m\}\]${PROMPT_SEPARATOR@P}${PROMPT_SEPARATOR_COLOR:+\[\e\0m\]${PROMPT_HIGHLIGHT:~}\[\e\]${PROMPT_HIGHLIGHT}m\]\[${PROMPT_COLOR:+\[\e\]${PROMPT_COLOR}m\}\]${PROMPT_DIR_COLOR:+\[\e\]${PROMPT_DIR_COLOR}m\}\]${PROMPT_DIRECTORY@P}${PROMPT_GIT_BRANCH:+${PROMPT_SEPARATOR:+\[\e\0m\]${PROMPT_SEPARATOR_COLOR:+\[\e\]${PROMPT_SEPARATOR_COLOR}m\}\]${PROMPT_SEPARATOR@P}${PROMPT_SEPARATOR_COLOR:+\[\e\0m\]${PROMPT_HIGHLIGHT:~}\[\e\]${PROMPT_HIGHLIGHT}m\]\[${PROMPT_COLOR:+\[\e\]${PROMPT_COLOR}m\}\]${PROMPT_DIR_COLOR:+\[\e\]${PROMPT_DIR_COLOR}m\}\]${PROMPT_GIT_COLOR:+\[\e\]${PROMPT_GIT_COLOR}m\}\]${PROMPT_GIT_BRANCH:~}\[\e\0m\]${PROMPT_END@P}}\${PROMPT_END:+\[\e\0m\]}\ \[\e\]133;k;start_suffix_kitty\a\]\[\e\]5 q\]\[\e\]2;\w\a\]\[\e\]133;k;end_suffix_kitty\a\]
54c57
< PATH=/home/tangc/.local/bin:/home/tangc/bin:/usr/local/bin:/usr/bin
---
> PATH=/home/tangc/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/.venv/bin:/home/tangc/.local/bin:/home/tangc/bin:/usr/local/bin:/usr/bin
```

图 10: 课后练习 · A1: printenv 前后快照与 diff, 可见 VIRTUAL_ENV、PS1 与 PATH 变化

```

in:/usr/local/bin:/usr/bin
(.venv) tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ which deactivate
deactivate ()
{
    if [ -n "${_OLD_VIRTUAL_PATH:-}" ]; then
        PATH="${_OLD_VIRTUAL_PATH:-}";
        export PATH;
        unset _OLD_VIRTUAL_PATH;
    fi;
    if [ -n "${_OLD_VIRTUAL_PYTHONHOME:-}" ]; then
        PYTHONHOME="${_OLD_VIRTUAL_PYTHONHOME:-}";
        export PYTHONHOME;
        unset _OLD_VIRTUAL_PYTHONHOME;
    fi;
    hash -r 2> /dev/null || true;
    if [ -n "${_OLD_VIRTUAL_PS1:-}" ]; then
        PS1="${_OLD_VIRTUAL_PS1:-}";
        export PS1;
        unset _OLD_VIRTUAL_PS1;
    fi;
    unset VIRTUAL_ENV;
    unset VIRTUAL_ENV_PROMPT;
    if [ ! "${1:-}" = "nondestructive" ]; then
        unset -f deactivate;
    fi
}
(.venv) tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ deactivate
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ echo "$PATH"
/home/tangc/.local/bin:/home/tangc/bin:/usr/local/bin:/usr/bin
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$

```

图 11: 课后练习 · A1: which deactivate 显示其为函数, deactivate 后 PATH 还原

5.2 A2. 建包 + 生成并检查 lockfile

```

cp -r ../q09 shipping-code-2 && cd shipping-code-2
rm -rf .venv # 复制时别带 venv (不可移植)
# 给 pyproject.toml 补 requires-python = ">=3.13", dependencies = ["requests>=2.31"]
pip install uv
uv sync # 建/复用 .venv 并真正安装
uv tree # 依赖树
head -40 uv.lock # 每个包有 name/version/hash

```

```

tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cp ../q09 shipping-code-2
cp: -r not specified; omitting directory '../q09'
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cp -r ../q09 shipping-code-2
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cd ./shipping-code-2/
tangc@fedora:~/../week3/ex/shipping-code-2$ python -m venv .venv && source .venv/bin/activate
(.venv) tangc@fedora:~/../week3/ex/shipping-code-2$ uv lock
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 2ms
(.venv) tangc@fedora:~/../week3/ex/shipping-code-2$ head -20 uv.lock
version = 1
revision = 3
requires-python = ">=3.14"

[[package]]
name = "greetlab-25020007111"
version = "0.1.0"
source = { editable = "." }
(.venv) tangc@fedora:~/../week3/ex/shipping-code-2$

```

图 12: 课后练习 · A2: 初次 cp -r ../q09 + uv lock, head -20 uv.lock 查看锁文件 (此时仅有项目自身一条)

```

~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-code-2/q09
tango@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cp -r ../q09 shipping-code-2
tango@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cd ./shipping-code-2/
tango@fedora:~/../week3/ex/shipping-code-2$ rm -rf ./q09/.venv
tango@fedora:~/../week3/ex/shipping-code-2$ python -m venv .venv && source .venv/bin/activate
(.venv) tango@fedora:~/../week3/ex/shipping-code-2$ uv lock
error: No `pyproject.toml` found in current directory or any parent directory
(.venv) tango@fedora:~/../week3/ex/shipping-code-2$ cd ./q09/
(.venv) tango@fedora:~/../ex/shipping-code-2/q09$ uv lock
Using CPython 3.14.7 interpreter at: /usr/bin/python
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 2ms
(.venv) tango@fedora:~/../ex/shipping-code-2/q09$ uv sync
warning: `VIRTUAL_ENV=/home/tango/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-code-2/.venv` does not
match the project environment path `.venv` and will be ignored; use `--active` to target the active environment instead
Using CPython 3.14.7 interpreter at: /usr/bin/python
Creating virtual environment at: .venv
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 1ms
Built greetlab-25020007111 @ file:///home/tango/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-co
Prepared 1 package in 2.27s
Installed 1 package in 2ms
+ greetlab-25020007111==0.1.0 (from file:///home/tango/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-
code-2/q09)
(.venv) tango@fedora:~/../ex/shipping-code-2/q09$ uv tree
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 1ms
greetlab-25020007111 v0.1.0
(.venv) tango@fedora:~/../ex/shipping-code-2/q09$ head -20 uv.lock
version = 1
revision = 3
requires-python = ">=3.14"

[[package]]
name = "greetlab-25020007111"
version = "0.1.0"
source = { editable = "." }
(.venv) tango@fedora:~/../ex/shipping-code-2/q09$ |

```

图 13: 课后练习 · A2: uv lock/uv sync 解析依赖并创建 venv, uv tree 与 uv.lock 检查

```

tango@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cp -r ../q09 shipping-code-2 && cd shipping-code-2
tango@fedora:~/../week3/ex/shipping-code-2$ ls
dist pyproject.toml src
tango@fedora:~/../week3/ex/shipping-code-2$ ls -a
. .. dist pyproject.toml src .venv
tango@fedora:~/../week3/ex/shipping-code-2$ rm -rf .venv
tango@fedora:~/../week3/ex/shipping-code-2$ uv sync
Using CPython 3.14.7 interpreter at: /usr/bin/python
Creating virtual environment at: .venv
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 4ms
Built greetlab-25020007111 @ file:///home/tango/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-co
Prepared 1 package in 814ms
Installed 1 package in 2ms
+ greetlab-25020007111==0.1.0 (from file:///home/tango/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/shipping-
code-2)
tango@fedora:~/../week3/ex/shipping-code-2$ uv tree
warning: No `requires-python` value found in the workspace. Defaulting to `>=3.14`.
Resolved 1 package in 1ms
greetlab-25020007111 v0.1.0
tango@fedora:~/../week3/ex/shipping-code-2$ head -20 uv.lock
version = 1
revision = 3
requires-python = ">=3.14"

[[package]]
name = "greetlab-25020007111"
version = "0.1.0"
source = { editable = "." }
tango@fedora:~/../week3/ex/shipping-code-2$

```

图 14: 课后练习 · A2: 先 rm -rf .venv 再 uv sync, head -20 uv.lock 查看锁定内容

uv lock 只解析、不安装；要真正安装用 uv sync。lockfile 把每层依赖的精确版本与哈希写死，保证可复现。

5.3 A3. 装 Docker, 用 docker compose 本地构建 Missing Semester 网站

```

sudo dnf install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
sudo systemctl enable --now docker
sudo usermod -aG docker "$USER"

```

```
git clone https://github.com/missing-semester/missing-semester.git
cd missing-semester
docker compose up          # 构建 Jekyll 站点并起服务, http://localhost:4000
```

```
tangc@fedora:~$ docker compose version
Docker Compose version 5.5.0
tangc@fedora:~$ sudo systemctl enable --now docker
tangc@fedora:~$ sudo usermod -aG docker "$USER"
tangc@fedora:~$ cd ~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ git clone https://github.com/missing-semester/missing-semester.git
Cloning into 'missing-semester'...
remote: Enumerating objects: 3465, done.
remote: Counting objects: 100% (250/250), done.
remote: Compressing objects: 100% (133/133), done.
remote: Total 3465 (delta 151), reused 117 (delta 117), pack-reused 3215 (from 2)
Receiving objects: 100% (3465/3465), 12.73 MiB | 4.37 MiB/s, done.
Resolving deltas: 100% (2101/2101), done.
tangc@fedora:~/../sys-dev-tools-course-2026/week3/ex$ cd missing-semester
```

图 15: 课后练习 · A3: docker compose version 确认 5.5.0, 启动 docker 并 clone missing-semester 仓库

```
> ~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/missing-semester
tangc@fedora:~/../week3/ex/missing-semester$ docker compose up
[+] up 0/1
❗ Image missing-semester:latest Pulling 3.8s
[+] Building 818.0s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 664B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 323B 0.0s
=> [internal] load metadata for docker.io/library/ruby:3.4-alpine3.21 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/ruby:3.4-alpine3.21@sha256:9fff0a06c84ede358c91ce95fc3917c3dd65f92d119f5e318567683b6fa9c063 0.1s
=> => resolve docker.io/library/ruby:3.4-alpine3.21@sha256:9fff0a06c84ede358c91ce95fc3917c3dd65f92d119f5e318567683b6fa9c063 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 8.75kB 0.0s
=> [2/6] RUN apk add --no-cache ruby-dev alpine-sdk 62.8s
=> [3/6] RUN mkdir /app 0.3s
=> [4/6] COPY Gemfile Gemfile.lock /app/ 0.0s
=> [5/6] WORKDIR /app 0.0s
=> [6/6] RUN bundle install 726.2s
=> exporting to image 28.2s
=> => exporting layers 24.6s
=> => exporting manifest sha256:b4a65530b160b4ed737c482aeda121aa9d9475919e495b31f08047268c58bf03 0.0s
=> => exporting config sha256:63f72043d5cd18058cd190c444fb3616cd95d08788d8226794fef98c3bdfef28 0.0s
=> => exporting attestation manifest sha256:a212ea0c72e784b11787bde6cea39dc193b8ad6342e334073e596c0bffe96af4 0.0s
=> => exporting manifest list sha256:4fbaf29d526b5cd2be04e6309350049ba06e50abb3a58f689f95276a2c775545 0.0s
=> => naming to docker.io/library/missing-semester:latest 0.0s
[+] up 3/3acking to docker.io/library/missing-semester:latest 3.5s
✓ Image missing-semester:latest Built 821.9s
✓ Network missing-semester_default Created 0.1s
✓ Container missing-semester-server-1 Created 0.2s
Attaching to server-1
server-1 | Could not locate Gemfile or .bundle/ directory
server-1 exited with code 10 (restarting)
server-1 | Could not locate Gemfile or .bundle/ directory
server-1 exited with code 10 (restarting)
server-1 | Could not locate Gemfile or .bundle/ directory
~/Documents/obsidian_vault/git_repo/sys-dev-tools-course-2026/week3/ex/missing-semester
```

图 16: 课后练习 · A3: 首次 docker compose up 构建 Ruby 镜像成功, 但容器报 Could not locate Gemfile 反复重启

```
tangc@fedora:~/../week3/ex/missing-semester$ cat docker-compose.yml
services:
  server:
    image: missing-semester:latest
    build:
      dockerfile: Dockerfile
      context: .
    ports:
      - 4000:4000
    volumes:
      - ./:app:z
    restart: on-failure
tangc@fedora:~/../week3/ex/missing-semester$ docker compose down
[+] down 2/2
✓ Container missing-semester-server-1 Removed 0.0s
✓ Network missing-semester_default Removed 0.1s
tangc@fedora:~/../week3/ex/missing-semester$ docker compose up
[+] up 2/2
✓ Network missing-semester_default Created 0.1s
✓ Container missing-semester-server-1 Created 0.1s
Attaching to server-1
server-1 | Configuration file: /app/_config.yml
server-1 | Source: /app
server-1 | Destination: /app/_site
server-1 | Incremental build: disabled. Enable with --incremental
server-1 | Generating...
server-1 | done in 1.165 seconds.
server-1 | Auto-regeneration: enabled for '/app'
server-1 | Server address: http://0.0.0.0:4000
server-1 | Server running... press ctrl-c to stop.
tangc@fedora:~/../week3/ex/missing-semester$ docker compose up
[+] up 1/1
✓ Container missing-semester-server-1 Running 0.0s
Attaching to server-1
```

图 17: 课后练习 · A3: 查看 docker-compose.yml, down 后重新 up, 容器正常启动并监听 4000

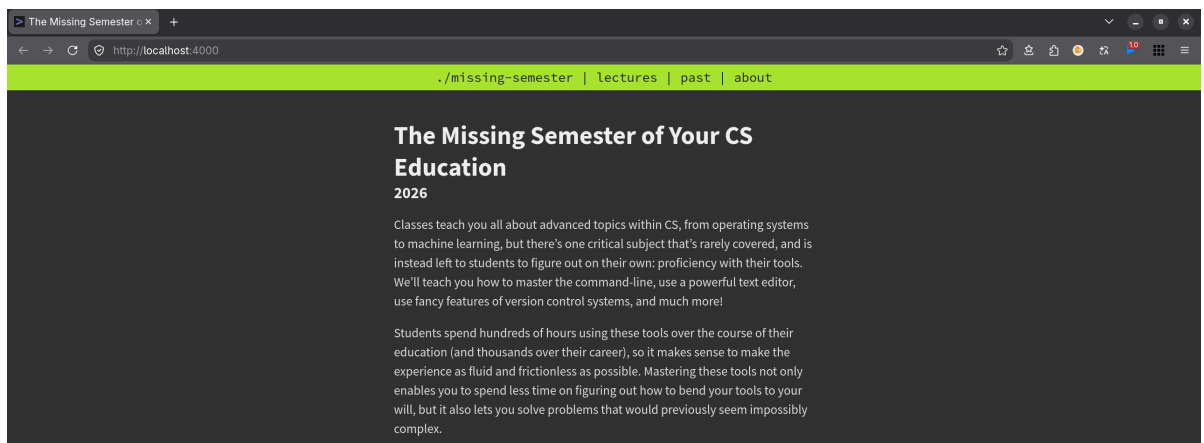


图 18: 课后练习 · A3: 浏览器打开 http://localhost:4000 显示 Missing Semester 首页



图 19: 课后练习 · A3: 站内页面 /2026/course-shell/ 正常渲染

5.4 A4. 写 Dockerfile + docker-compose.yml (应用 + Redis 缓存)

```
# app.py (Flask + Redis 访问计数)
import os
from flask import Flask
import redis
app = Flask(__name__)
r = redis.Redis(host=os.environ.get("REDIS_HOST", "localhost"),
                port=6379, decode_responses=True)
@app.route("/")
def home():
    return f"访问次数: {r.incr('hits')}}"
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
# Dockerfile
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
CMD ["python", "app.py"]
```

```
# docker-compose.yml
services:
  web:
    build: .
    ports:
      - "5000:5000"
    environment:
      - REDIS_HOST=cache # 服务名即内网 DNS
    depends_on:
      - cache
  cache:
    image: redis:7-alpine
    volumes:
      - redis_data:/data
volumes:
  redis_data:
```

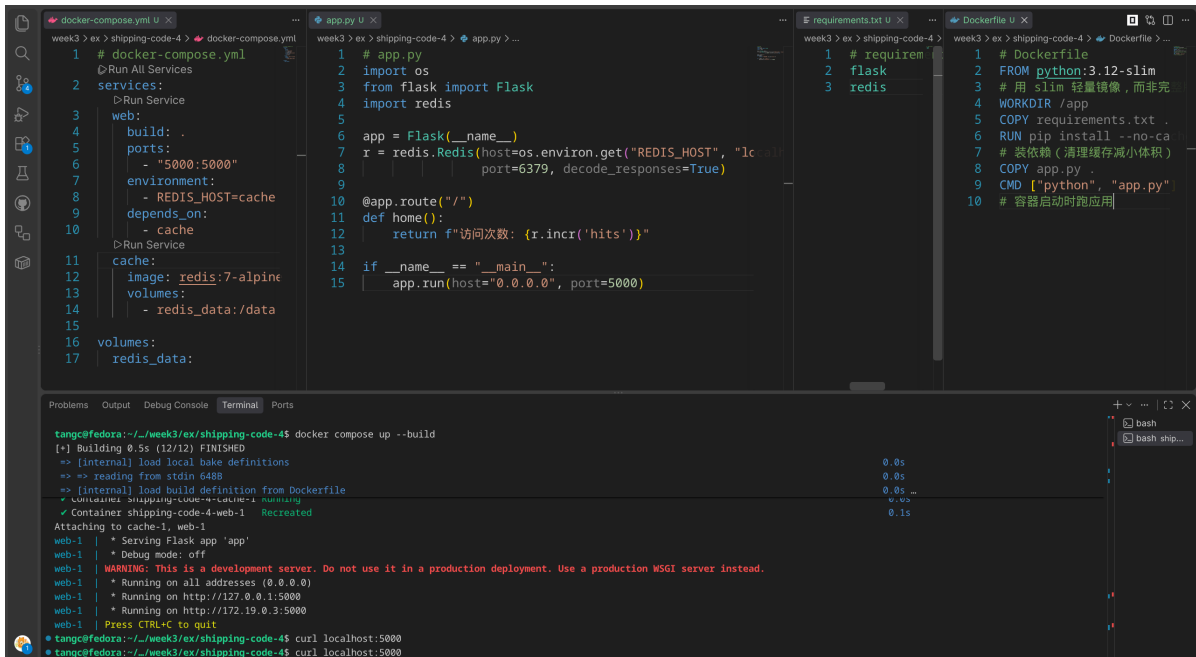
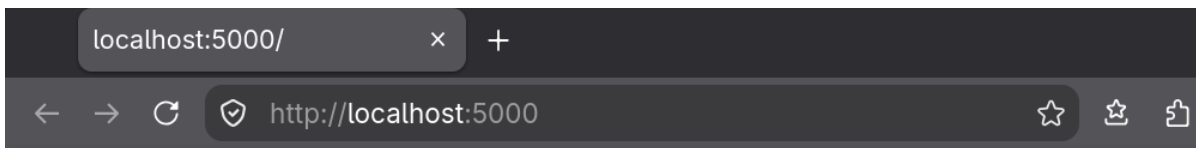


图 20: 课后练习 · A4: 四个文件 (Dockerfile / app.py / requirements.txt / docker-compose.yml) 与 docker compose up --build 运行



访问次数: 6

图 21: 课后练习 · A4: 浏览器访问 localhost:5000 显示 “访问次数: 6”，Redis 计数生效

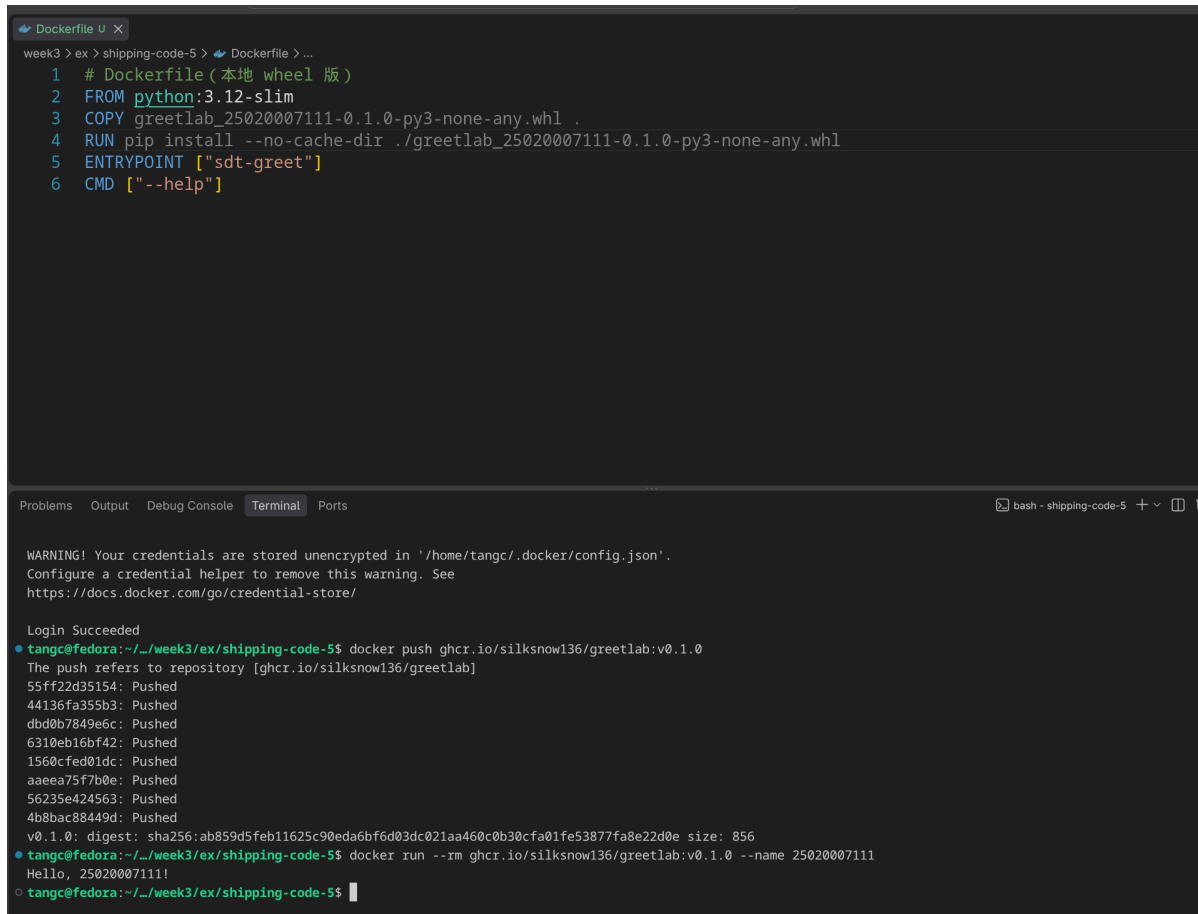
5.5 A5. 发布到 TestPyPI + 打 Docker 镜像推到 ghcr.io

```
python -m build                # 产出 wheel
# 方式 A: twine upload -r testpypi dist/* (需 TestPyPI 账号 2FA + token)
# 方式 B (TestPyPI 受阻时的等效做法) : Docker 镜像装本地 wheel
docker build -t ghcr.io/silksnow136/greetlab:v0.1.0 .
echo "ghp_XXX" | docker login ghcr.io -u silksnow136 --password-stdin
docker push ghcr.io/silksnow136/greetlab:v0.1.0
docker run --rm ghcr.io/silksnow136/greetlab:v0.1.0 --name 25020007111
```

本地 wheel 版 Dockerfile:

```
FROM python:3.12-slim
COPY greetlab_25020007111-0.1.0-py3-none-any.whl .
RUN pip install --no-cache-dir ./greetlab_25020007111-0.1.0-py3-none-any.whl
```

```
ENTRYPOINT ["sdt-greet"]
CMD ["--help"]
```



```
Dockerfile U X
week3 > ex > shipping-code-5 > Dockerfile > ...
1 # Dockerfile (本地 wheel 版)
2 FROM python:3.12-slim
3 COPY greetlab_25020007111-0.1.0-py3-none-any.whl .
4 RUN pip install --no-cache-dir ./greetlab_25020007111-0.1.0-py3-none-any.whl
5 ENTRYPOINT ["sdt-greet"]
6 CMD ["--help"]

Problems Output Debug Console Terminal Ports
WARNING! Your credentials are stored unencrypted in '/home/tangc/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
tanc@fedora:~/week3/ex/shipping-code-5$ docker push ghcr.io/silksnow136/greetlab:v0.1.0
The push refers to repository [ghcr.io/silksnow136/greetlab]
55ff22d35154: Pushed
44136fa355b3: Pushed
dbd0b7849e6c: Pushed
6310eb16bf42: Pushed
1560cfed01dc: Pushed
aabee75f7b0e: Pushed
56235e424563: Pushed
4b8bac88449d: Pushed
v0.1.0: digest: sha256:ab859d5feb11625c90eda6bf6d03dc021aa460c0b30cfa01fe53877fa8e22d0e size: 856
tanc@fedora:~/week3/ex/shipping-code-5$ docker run --rm ghcr.io/silksnow136/greetlab:v0.1.0 --name 25020007111
Hello, 25020007111!
tanc@fedora:~/week3/ex/shipping-code-5$
```

图 22: 课后练习 · A5: Dockerfile (本地 wheel 版); docker login 成功后 push 到 ghcr.io, 并 docker run ... --name 25020007111 输出 Hello

5.6 B1. 用智能体浏览陌生代码库 (理解 opencode 的安全机制)

让智能体浏览 opencode 仓库, 解释权限请求拦截、权限模式配置与沙箱机制, 并要求给出源码文件路径与行号。

请浏览当前仓库 (opencode), 解释它的安全 / 权限机制是怎么实现的:

1. 用户确认工具调用 (权限请求) 是在哪一步、怎么拦截的?
2. 有哪些权限模式 (只读、自动批准等)? 分别在哪个文件配置?
3. 沙箱 / 隔离是怎么做的? 是否依赖容器或系统机制?

请给出关键源码文件路径和行号, 不要贴大段代码。

智能体会话记录 (share): <https://opncd.ai/share/FJj9fIkV>

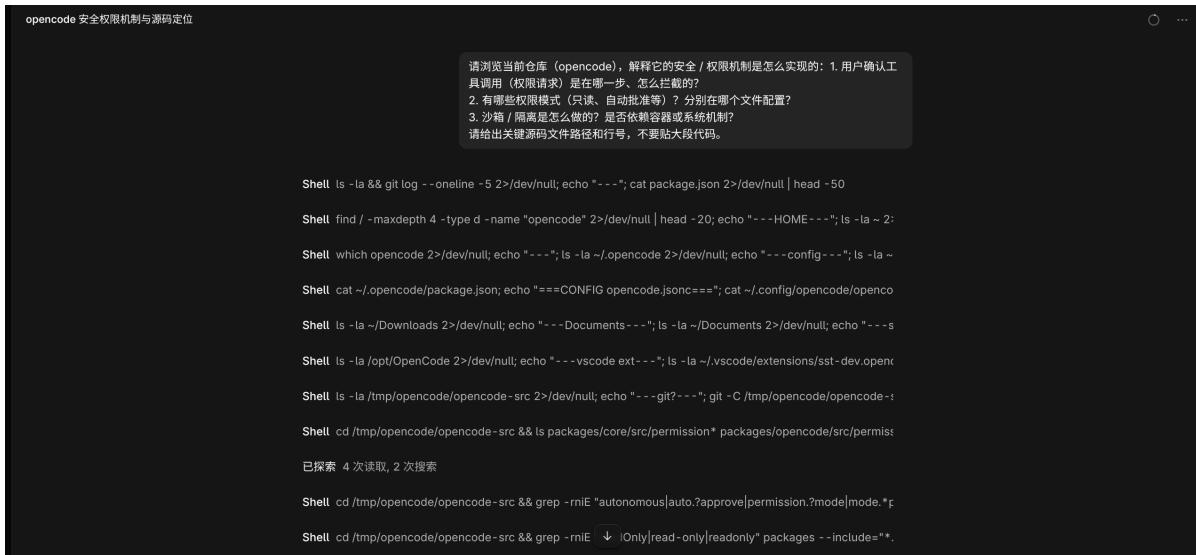


图 23: 课后练习 · B1: 智能体在 opencode 仓库中反复 grep 权限相关源码定位实现

5.7 B2. 从零氛围编程一个小应用（不手写一行代码）

给智能体一个 C++ 目标，让它自行用 CMake 组织项目、写代码、编译、运行、迭代。

用 C++ 写一个命令行小工具 csv2md：读取 CSV 文件，输出 Markdown 表格。

要求：用 CMake 组织项目生成单个可执行文件 csv2md；正确处理引号包裹、含逗号的字段；自己写样例 CSV，用 g++/cmake 编译并运行验证输出；自己迭代直到能跑通。

```
cmake -S . -B build && cmake --build build
./build/csv2md sample.csv
```

智能体会话记录 (share): <https://opncd.ai/share/b4J42S5E>



图 24: 课后练习 · B2: 智能体自动写 CMakeLists.txt / main.cpp / sample.csv，构建并运行，还自行改代码迭代

```
tangc@fedora:~/Documents/codingCoding/codeRef/vibe$ cmake -S . -B build && cmake --build build
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/tangc/Documents/codingCoding/codeRef/vibe/build
[100%] Built target csv2md
tangc@fedora:~/Documents/codingCoding/codeRef/vibe$ ls
build CMakeLists.txt sample.csv src
tangc@fedora:~/Documents/codingCoding/codeRef/vibe$ cat sample.csv && ./build/csv2md sample.csv
Name,Role,Note,Score
Alice,Engineer,"Loves ""parsing"" and commas, obviously",95
Bob,Designer,"Handles pipes | too",88
Carol,"Data, Analyst","Quoted ""field"" with, many commas",77
| Name | Role | Note | Score |
| --- | --- | --- | --- |
| Alice | Engineer | Loves "parsing" and commas, obviously | 95 |
| Bob | Designer | Handles pipes \| too | 88 |
| Carol | Data, Analyst | Quoted "field" with, many commas | 77 |
tangc@fedora:~/Documents/codingCoding/codeRef/vibe$
```

图 25: 课后练习 · B2: csv2md 正确处理引号、逗号、竖线等边界，输出 Markdown 表格

5.8 B3. AGENTS.md / Skill / 子智能体 (何时用哪个)

在 week3/ex/agent-coding-3/q09 下建 AGENTS.md、Skill(.opencode/skills/python-code-check/SKILL.md) 与子智能体 (.opencode/agent/code-reviewer.md)，重启 opencode 后分别触发验证。

```
# AGENTS.md (项目根，永远加载)
# greetlab 项目约定
- 改完代码必须运行 `PYTHONPATH=src pytest` 并通过。
- 提交前运行 `ruff check .`，不得留下未使用 import。
- 不要修改 `pyproject.toml` 的 name/version。
- 行为约定：`--name` 只含空白字符时必须以退出码 2 结束。
```

```
# .opencode/skills/python-code-check/SKILL.md
---
name: python-code-check
description: Use when checking changed Python files in greetlab. 对改动过的 .py 跑 ruff 与
  pytest，给出问题清单。
---
1. git diff --name-only 找出改动的 .py 文件；
2. 对每个文件跑 ruff check；
3. 跑 PYTHONPATH=src pytest；
4. 汇总问题（文件:行号 + 建议），不改文件。
```

```
# .opencode/agent/code-reviewer.md
---
description: 审查 greetlab 未提交改动，输出分级评审，不改文件。
mode: subagent
permission:
  edit: deny
---
审查未提交改动，重点看：空白 name 是否真的以退出码 2 结束、是否误改了 test/pyproject。
按 Blocking / Suggestion / Nit 三级输出中文评审意见。
```

三种手段的区别：AGENTS.md 启动时全量常驻；Skill 只注入 name+description、按需加载正文；子智能体独立上下文、由主智能体通过 task 调用。改完须重启 opencode。

智能体会话记录 (share): <https://opncd.ai/share/5HOMEJI3>

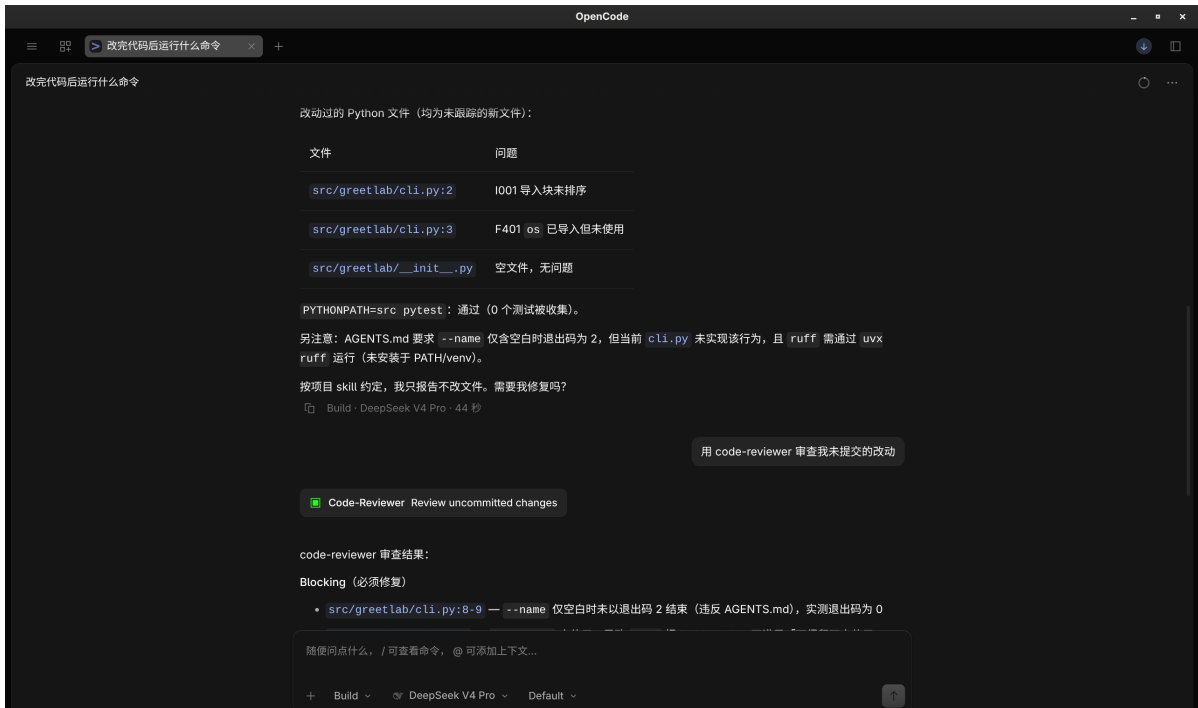


图 26: 课后练习·B3: 触发 Skill 得到改动文件清单(I001/F401)与 pytest 结果;再让 code-reviewer 子智能体审查, 输出 Blocking 级意见

5.9 B4. 用智能体完成 Markdown 无序列表正则替换（不直接编辑文件）

目标: 把 code-quality.md 里行首的 - 列表标记替换成 * , 但保留日期/范围/选项里的 -。
约束智能体不得直接编辑文件, 改用 sed。

把 code-quality.md 里行首的无序列表标记 ``-`` 替换为 ``*``。

约束: 不要直接编辑文件; 用命令行工具 (sed) 加正则完成;

日期(2026-01-14)、范围(a-f)、选项(-q) 等其它位置的 ``-`` 必须保留。

完成后给出你用的 sed 命令, 并贴 git diff 让我复核。

```
sed -i 's/^- /* /' code-quality-after.md # ^ 锚定行首, 只改列表标记
git diff --no-index --stat code-quality.md code-quality-after.md
```

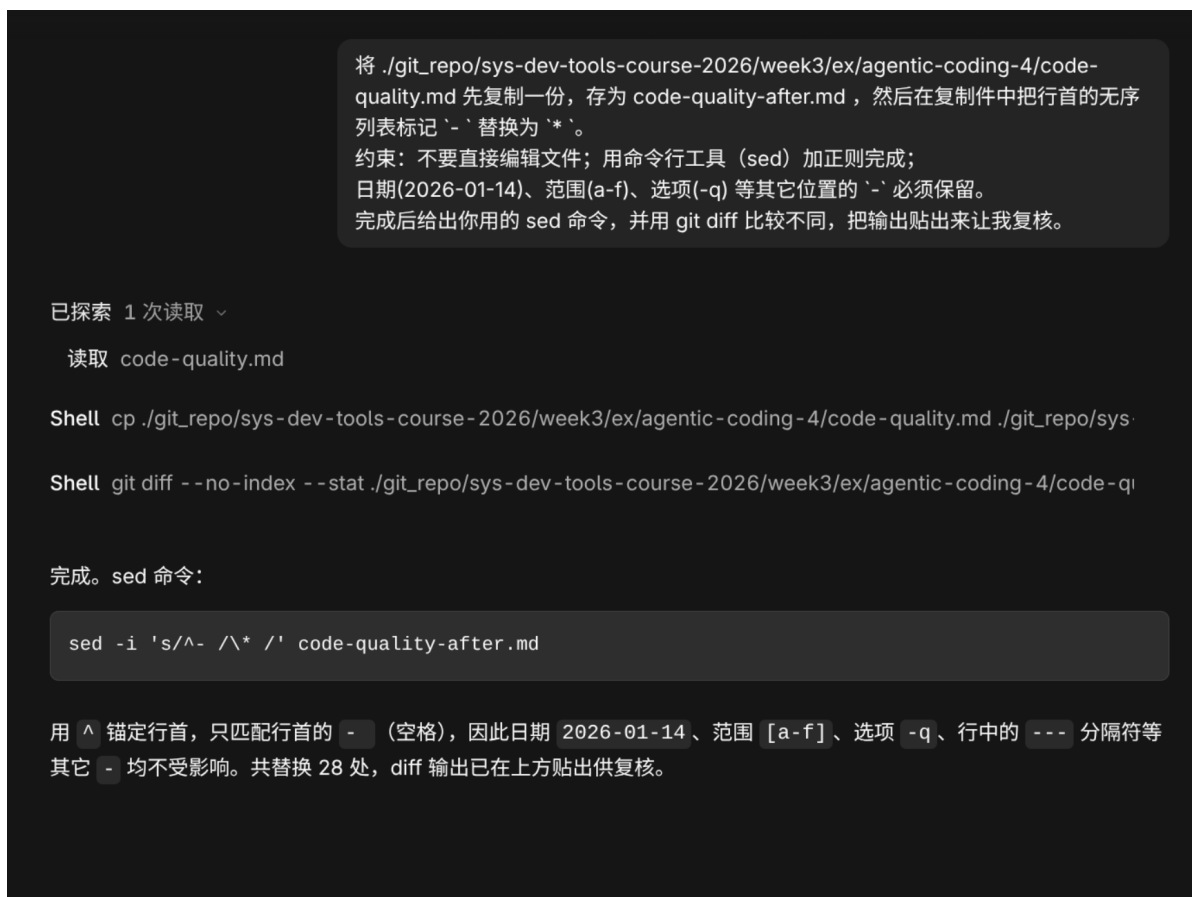


图 27: 课后练习 · B4: 智能体先复制文件再用 `sed -i 's/^- /\* /'` 替换行首列表标记 (共 28 处), 未直接编辑, 日期/范围/选项中的 `-` 保留

智能体会话记录 (share): <https://opncl.ai/share/ySJqhzoD>

5.10 C1. 浏览知名项目源码, 找出几类“值得写”的注释

选 C++ 项目 `nlohmann/json` 与 `fmt`, 用 `rg` 定位注释类型。实测结果:

```
git clone https://github.com/nlohmann/json.git && cd json
rg -n "TODO|FIXME|HACK" include/ src/ # 带上下文的待办
rg -n "https://|RFC|paper|wg21" include/ # 外部引用
rg -n -i "we use .* instead|deliberately|do not use" include/ # why-not
```

- 外部引用: `json.hpp:25` 挂 issue、`json.hpp:5026` 引 RFC 6902、`lexer.hpp:335` 对应 RFC 8259、`to_chars.hpp:345` 锚定 Dragonbox 论文。
- why-not / 血泪教训: `json.hpp:5459` 的 `// do not remove the space after '<'` 是编译器 workaround; `fmt/core.h:651` 的 `intentionally not constexpr` 是刻意设计。
- TODO: `json.hpp:1268` 是配合 `cppcheck` 的 `// cppcheck-suppress ... TODO check`; `fmt` 源码里 `rg TODO` 竟零命中。

心得: 注释价值排序为 `why-not` > 外部引用 > 带上下文 TODO > 空 TODO。删掉会让人踩坑的, 才叫“值得写”的注释。

5.11 C2. 好提交信息 vs 弱提交信息（体会补背景的费劲）

用 `git log --oneline -20` 对比 `fmt` 与 `json`:

- 好提交: `json ff80ed329` “Speed up `dump()`, and keep it from overflowing the stack” 正文讲清问题、方案、基准（4.2x/2.7x）与影响。
- 弱提交: `fmt 14174c82` “Add braces for readability”、`json 3e2de3222` “Remove unused `iomanip include`”，正文为空。

按“问题 → 解决方案 → 影响”重写弱提交,例如 `fix crash` 可改为 `Fix use-after-free in parse when in` 正文说明空输入导致 UAF、提前校验 `input.empty()`、代价可忽略。

心得: 弱提交把反推成本转嫁给了未来——好提交当下多花 30 秒, 弱提交让未来每个读它的人多花几分钟 `git show` 翻代码猜动机。

5.12 C3. 评估一个未解决的 issue

用 GitHub REST API 取 `nlohmann/json` 的 good first issue:

```
curl -sS "https://api.github.com/repos/nlohmann/json/issues?labels=good%20first%20issue&state=open&per_page=10" \
| jq -r '.[0] | "#\(.number) \(.title)'"
```

- #5318(极高): 贴最小复现 + 定位到 `binary_writer.hpp:403-407` + 环境 `g++ 13.3.0/Ubuntu 24.04` + 分配次数表 + 建议修法, 维护者可直接开工。
- #5317 (极高): 引 RFC 8949 §3.2.3、给接受/拒绝对照表与复现, 标准到可当模板。
- #3885 (低): `Minimal code example`: Na、期望含糊, 维护者被逼追问 3+ 轮才定位。

心得: issue 质量直接决定维护者时间花在哪——好 issue 让维护者直接改代码, 坏 issue 让维护者先当侦探。落点是一句话: 维护者看完能不能立刻开工?

6 提交记录

课程要求将每次报告、练习、代码上传到个人 GitHub 并保留 `commit` 记录（禁止单次全量提交）。本报告采用分批提交: 笔记 / 报告源码 / 报告 PDF / q09~q12 / 课后练习各占若干次提交。

- 仓库链接: <https://github.com/silksnow/sys-dev-tools-course-2026>
- 提交记录截图:

Commits		
main	All users	
Commits on Sep 11, 2026		
docs: 重新编译第2周实验报告PDF(36页)	silksnow136 committed 6 minutes ago	9dcfb787
docs: 更新第2周实验报告源码(补充性能4/5与tmux截图)	silksnow136 committed 6 minutes ago	eb95c8e
docs: 添加第3周实验题笔记(打包发布/智能体编程/不止于代码/PyTorch)	silksnow136 committed 6 minutes ago	7f7c3d4
docs: 更新第2周笔记(补充Wireshark抓包与tmux详解)	silksnow136 committed 7 minutes ago	d8d52b9
docs: 更新第1周笔记(补充位置参数 \$@/\$* 说明)	silksnow136 committed 7 minutes ago	f989574
Commits on Sep 10, 2026		
chore: 添加 .gitattributes 统一 LF 换行符	silksnow136 committed 18 hours ago	ab360c4
Commits on Sep 7, 2026		
docs: 添加第2周实验报告PDF(含Q5-Q8与课后练习截图)	silksnow136 and silksnow committed 4 days ago	9bde08a
docs: 添加第2周实验报告源码(LaTeX+xelatex编译)	silksnow136 and silksnow committed 4 days ago	65dd6d7
docs: 添加第2周实验题笔记(课后练习/Python速查C++对照/目录跳转)	silksnow136 and silksnow committed 4 days ago	d74ec3d
feat: 第2周课后练习 ex(shellcheck调试3/fib.py/sorts.py/pycallgraph调用图)	silksnow136 and silksnow committed 4 days ago	39a4741
Commits on Sep 4, 2026		
feat: 第2周 q08 测量并优化慢速词频排序	silksnow136 and silksnow committed last week	459b744
feat: 第2周 q06, q07 (语义重构与本地开发反馈, 用调试器定位归并排序缺陷) 完成。	silksnow136 and silksnow committed last week	754d7d5
feat: 第2周实验 q05 可清理后台任务(任务控制/信号)	silksnow136 and silksnow committed last week	9e8950c

图 28: GitHub commit 记录 (含第 3 周笔记、第 1/2 周报告与笔记更新等分批提交)